

Automation with Gen AI

# Practical Capstone Projects

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10 Projects · Voice · Multimodal · Agent · Real-World Tools

These projects go beyond standard Gen AI tutorials. Each one solves a real problem a non-technical person would immediately recognise and want to use. They cover voice, image understanding, agent-style automation, and practical tools that are genuinely portfolio-worthy.

## VOICE & AUDIO

### Project 01

## Mock Interview Coach

*"Speak your answer, hear your feedback."*

### BUILD

Build a Streamlit app where a user selects an interview type (technical, HR, or management), receives a question, records or types their answer, and gets AI-evaluated feedback on clarity, relevance, and confidence. The app then reads the feedback aloud using ElevenLabs.

### USE

**Claude API + ElevenLabs API + Streamlit + python-dotenv**

### EXPECTED OUTPUT

**A working interview simulation with at least 10 questions per category, AI scoring per answer with written and spoken feedback, and a downloadable session report with scores.**

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### Project 02

## Multilingual Document Narrator

*"Upload any PDF, hear its summary in any language."*

### BUILD

Build a Streamlit app where a user uploads a PDF, selects a target language from at least 8 options, and the app summarises the document using AI and converts the summary to natural-sounding spoken audio in the selected language using ElevenLabs.

### USE

**Claude API + ElevenLabs API + Streamlit + pypdf + python-dotenv**

### EXPECTED OUTPUT

**A working app that produces an accurate summary and a downloadable MP3 audio file in the chosen language, with graceful handling of PDFs over 10 pages and unsupported language fallback.**

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## MULTIMODAL (IMAGE UNDERSTANDING)

### Project 03

## Product Listing Generator

*"Photograph your product, get a marketplace listing ready to publish."*

### BUILD

Build a Streamlit app where a user uploads one or more product photos. The AI reads the images and generates a complete e-commerce product listing: title, 5 bullet point features, full description, and suggested keywords. User selects the target platform style (Amazon, Flipkart, or general).

### USE

**GPT-4o Vision API + Streamlit + python-dotenv**

#### EXPECTED OUTPUT

A listing generated within 30 seconds per product, all four components present (title, bullets, description, keywords), platform style visibly affecting output, and a one-click copy or download of the complete listing.

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#### Project 04

## Whiteboard To Documentation Converter

*"Photograph a whiteboard, get clean structured documentation."*

#### BUILD

Build a Streamlit app where a user uploads a photo of a whiteboard containing notes, diagrams, or flowcharts. The AI reads the image and converts the contents into structured documentation with headings, bullet points, and a plain-English description of any diagrams present.

#### USE

[GPT-4o Vision API](#) + [Streamlit](#) + [python-dotenv](#)

#### EXPECTED OUTPUT

Structured output that correctly identifies and separates text from diagram content, exported as a downloadable PDF, with a confidence note where the image is unclear or partially readable.

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#### Project 05

## Food Nutrition Analyser

*"Photograph your meal or food label, know what you are eating."*

#### BUILD

Build a Streamlit app where a user uploads a photo of a meal or a food product label. The AI estimates nutritional content (calories, protein, carbs, fat), flags common allergens, and suggests one healthier alternative for the main ingredient.

#### USE

[GPT-4o Vision API](#) + [Streamlit](#) + [python-dotenv](#)

#### EXPECTED OUTPUT

A structured nutrition panel displayed in the app, allergen flags highlighted visually, the healthier alternative suggestion with a brief reason, and a disclaimer that the output is an estimate and not medical advice.

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### AGENT-STYLE PROJECTS

#### Project 06

## Personalised News Briefing Agent

*"Define your interests once, get a curated briefing every morning."*

#### BUILD

Build a system where a user configures their topics of interest and RSS feed or URL sources. An n8n cron job runs daily, fetches content from those sources, the AI filters only relevant articles and generates a personalised briefing with summaries. The briefing is delivered by email every morning.

USE

**n8n + Claude API + Gmail + python-dotenv**

EXPECTED OUTPUT

**A fully automated daily email briefing with at least 5 relevant article summaries, each with a headline, 2-line summary, and source link, delivered at a configured time with no manual trigger required.**

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#### Project 07

## Github Repo Explainer

*"Paste any GitHub URL, understand the project in 60 seconds."*

BUILD

Build a Streamlit app where a user pastes a public GitHub repository URL. The app fetches the README, file structure, and key source files using the GitHub API. The AI reads all of this and produces a plain-English explanation of what the project does, how it is structured, and how to get started.

USE

**Claude API + GitHub API + Streamlit + python-dotenv**

EXPECTED OUTPUT

**A structured explanation with four sections: what it does, how it is built, how to get started, and known limitations — all generated from live repo data, not from Claude's training knowledge, with a download option.**

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### PRACTICAL REAL-WORLD TOOLS

#### Project 08

## Salary Negotiation Coach

*"Paste your offer letter, get a negotiation strategy and a ready-to-send counter email."*

BUILD

Build a Streamlit app where a user pastes a job offer letter and their target salary. The AI analyses the offer, identifies negotiation levers (base pay, bonus, equity, benefits), and generates a negotiation strategy with a professional counter-offer email ready to send.

USE

**Claude API + Streamlit + python-dotenv**

EXPECTED OUTPUT

**A structured negotiation brief with identified levers, a recommended counter-offer range with reasoning, and a professional email draft — all downloadable as a PDF, with a tone selector (assertive / balanced / conservative).**

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## Project 09

# College Application Essay Reviewer

*"Paste your personal statement, get paragraph-level feedback before you submit."*

### BUILD

Build a Streamlit app where a student pastes their personal statement essay and the prompt it is answering. The AI evaluates it against four criteria (structure, specificity, authenticity, relevance to prompt), scores each criterion, and gives paragraph-by-paragraph improvement suggestions.

### USE

[Claude API](#) + [Streamlit](#) + [python-dotenv](#)

### EXPECTED OUTPUT

**A score out of 100 with breakdown per criterion, specific paragraph-level comments (not generic advice), a revised version of the weakest paragraph as an example, and a downloadable marked-up review PDF.**

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## Project 10

# Api Documentation Generator

*"Paste your Python code, get complete professional documentation instantly."*

### BUILD

Build a Streamlit app where a developer pastes a Python script or set of functions. The AI reads the code and generates complete API documentation: function descriptions, parameter names and types, return values, usage examples, and edge case notes. Export as markdown.

### USE

[Claude API](#) + [Streamlit](#) + [python-dotenv](#)

### EXPECTED OUTPUT

**Complete documentation for every function detected in the pasted code, formatted as standard API docs with sections for description, parameters, returns, and example usage, downloadable as a markdown file and rendered in the app.**

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